SIMATS ENGINEERING

Lab Programs

COURSE CODE: ECA1009 **COURSE NAME: Microprocessors & Microcontrollers**

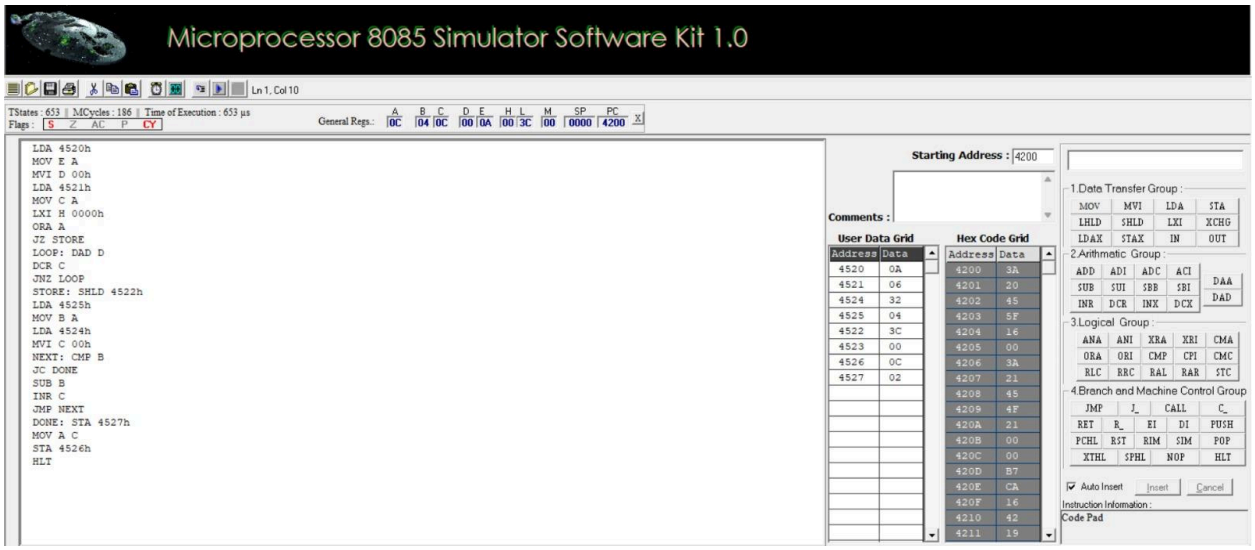
Program C – Implementation and Testing of 8-Bit Multiplication and Division Using Intel 8-Bit Microprocessor Assembly Language Programming

Aim: To implement and test 8-bit multiplication and division using Intel 8085 assembly language programming.

Algorithm:

1. Load the dividend and divisor or multiplicand and multiplier from the specified memory locations.
2. Initialize the registers required for the arithmetic operation.
3. Perform multiplication using repeated addition or division using repeated subtraction.
4. Repeat the operation until the required count or condition is satisfied.
5. Store the result and any remainder in the specified memory locations.
6. Halt the program after completing the operation.

Execute:



Result: The 8-bit multiplication and division program was executed successfully in the 8085 simulator, and the results were stored in the specified memory locations.